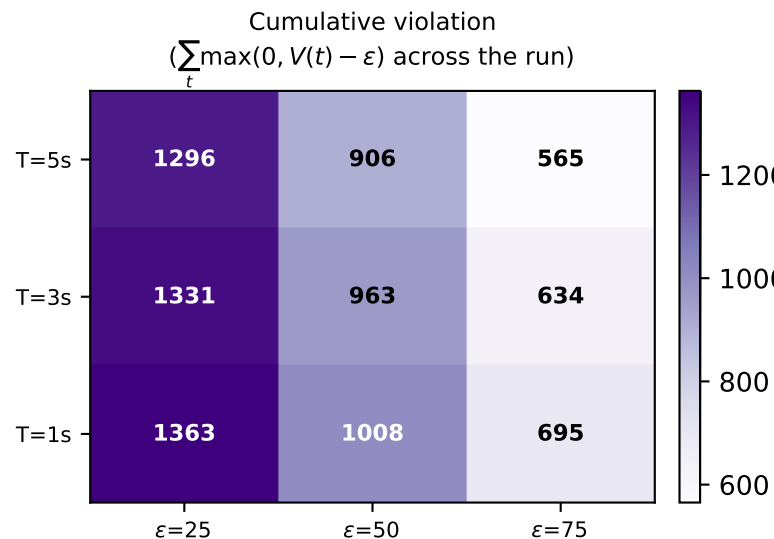
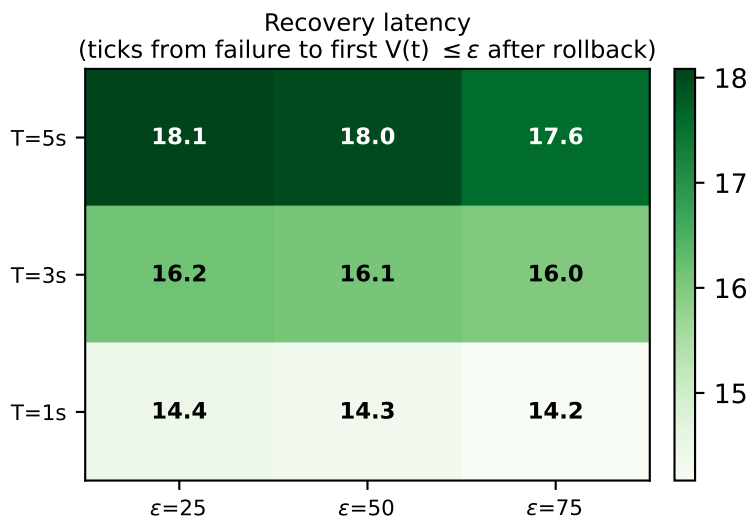
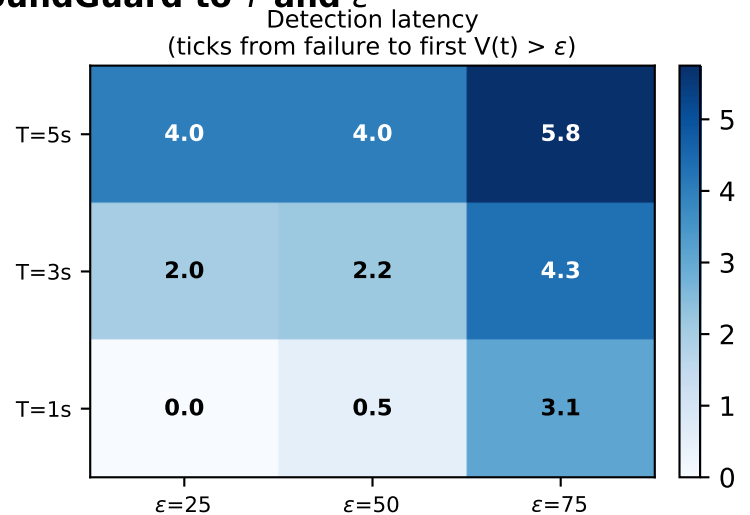


Detection sensitivity of BoundGuard to T and ε



Each cell in the four heatmaps reports the mean of the listed metric over $N = 12$ recovery runs (4 background-load scenarios $\times$ 3 repetitions, drawn from `\texttt{results/bounded\_sweep/run\_s*\_rep*.csv}`). The per-tick $v(\tau)$ column is fixed; only the windowed $V(t)$ computation is replayed for each (T, ε) combination, so the comparison is apples-to-apples and isolates the effect of the two detector knobs.

- **False triggers (top left).** The healthy phase 1 of the recovery scenario has $v(\tau) \approx 0.07$, so the windowed $V(t)$ never crosses any $\varepsilon \geq 25$ -- false-positive count is 0 in every cell on this grid. BoundGuard's two-stage rule (window-then-threshold) is therefore robust against single-tick jitter for any sensible threshold; the failure mode "too-low ε pollutes steady state" only kicks in below $\varepsilon \approx 1$.
- **Detection latency (top right).** Holding ε fixed, increasing T raises detection latency by exactly the additional samples the window has to accumulate before the test fires ($T - 1$ ticks of slack). Holding T fixed, raising ε from 25 to 75 also delays detection because $V(t)$ must climb further before crossing. Both effects are monotonic and predictable, matching the intuition that a more conservative detector (large T or large ε) reacts later.
- **Recovery latency (bottom left).** The recovery phase begins from the moment of failure injection, so longer detection delays push the whole recovery later. The bottom-left cell ($T = 1s, \varepsilon = 25$) is the fastest end-to-end recovery; the top-right cell